Certified Hole Energies and Stable Annular Gluing for the Critical XY Spin Field

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Abstract

We examine two analytic interfaces in the October 5 manuscript on the critical planar XY spin field, without independently certifying its critical-height input or its complete lattice-to-field conclusion. For separated circular holes, an exact Green boundary identity and an explicit annular Neumann flux give a one-sided quadratic error enclosure for the prescribed-period minimum energy, including unequal radii and general fixed real charges. This sharpens the source's stated linear radius remainder. For positive, nonstationary annular kernels on varying probability spaces, the exact norm of the normalization projection gives stability for average loss below $\sqrt{5}-2$, instead of the source's $1/64$, with an explicit finite-depth density estimate. We also give the weak residual and hypercircle certificates needed to use approximate energies without losing boundary or period information. Classical Hilbert-space projection, harmonic Fourier series, and positive-kernel arguments are credited as the underlying tools. Collision control, critical stiffness, microscopic core cancellation, and field tightness remain distinct source dependencies. Contribution assessment: C3/5, notable source-specific improvements to the hole-energy remainder and kernel-loss domain, with no independent global XY scaling-limit claim.

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1 Normalization and the exact scope of the comparison

The source [1] concerns the nearest-neighbor cosine XY model in a square, with every boundary angle fixed to zero and incident edges retained. Each unoriented edge is counted once. Its critical parameter is defined through the free-box mass, not by a choice of field amplitude. Write $D = (-1,1)^2$, and use

$$G_D = (-\Delta_D)^{-1}, \qquad G_D(z,w) = \frac{1}{2\pi}\log\frac{1}{|z-w|} + R_D(z,w), \qquad a = 8\pi, \quad K_* = \frac{a}{4\pi^2} = \frac{2}{\pi}. \tag{1}$$

The value $a = 8\pi$ is an imported critical-height assertion in the source. Our deterministic statements below do not prove it.

For the source's unscaled discrete Laplacian L_n, put $G_n = L_n^{-1}$ and $a_n = \mathbb{E}e^{i\theta_0} = \mathbb{E}\cos\theta_0 > 0$. The claimed normalized field is

$$S_n = \frac{1}{n^2}\sum_{x\in D_n^\circ} a_n^{-1}\exp\!\left(\frac{G_n(x,x) - G_n(0,0)}{2K_*}\right) e^{i\theta_x}\delta_x. \tag{2}$$

Its proposed limit is full-variance Wick-normalized imaginary chaos $:e^{i\Phi/\sqrt{K_*}}:$ for a Dirichlet GFF with covariance G_D. Thus its mean is one, and opposite charges have two-point kernel $\exp(G_D/K_*)$, with local exponent $1/(2\pi K_*) = 1/4$. There is no neutrality restriction for this fixed-angle boundary problem. The source does not require an asymptotic formula for a_n.

The height representation has spacing 2π. Its correction problem has *free* boundary heights modulo constants, despite the fixed spin angles. If ω_0 is a closed one-form with periods $2\pi l_i$, the relevant energy is

$$\mathcal{M}_U(\omega_0) = \inf_{g\in H^1(U)}\int_U |\omega_0 + \nabla g|^2, \qquad \text{sector limit } \exp[-\mathcal{M}_U/(2a)]. \tag{3}$$

Here U is a fixed connected Lipschitz domain with finitely many holes. The period data cannot be discarded. The coefficient in this height energy is $1/(2a)$, not $K_*/2$ applied to the same one-form.

1.1 Proof-dependency map

All seven scientific source files were read literally in bounded displays. Reading them is not an independent proof audit of their cited inputs. The following map records what must be preserved when substituting a local estimate.

Heights, source Section 2. Fourier currents become affine integer heights. Every cut component quotients its own constant; restoration sums relative integer translations. Gaussian pin interactions use a determinant and a Hilbert–Schmidt cross-covariance, not just an operator-norm bound. Critical Gaussian height limits and the stiffness in (1) are imported here.

Fixed sectors, Section 3. Free-boundary correction energies, tilted comparisons, cell penalties, and guarded cuts establish the stated fixed-hole sector limit. The localization argument combines exponential operator decay with a singular-value tail. Restoration errors are compared as partition ratios; a bound uniform over all conditional microscopic boundary data is not assumed.

Annuli and kernels, Sections 4–5. Confined constants, their subsequent release, and large annular modulus give average kernel losses. Conditioning is on full guard data. The exterior is one joint kernel, not a product of independent exterior pieces. Section 4 below replaces only the abstract positive-kernel stability lemma.

Correlations, Section 6. Green energies outside holes and identical unit-charge nests are compared with the center observable. The microscopic nest ratio cancels against a_n; it is not evaluated asymptotically. Section 3 improves only the continuum circular-hole error.

Fields, Section 7. A separate coarse lattice two-point bound, Lee–Yang moment estimates, collision removal, moment determinacy, and compactness produce the claimed full-sequence H^{-3}_{loc} limit. Separated-point formulas alone do not establish this step.

In particular, a sharper continuum exponent cannot simply be substituted for the source's coarse lattice bound. The measures in (2) are atomic; no stronger low-order Sobolev convergence is asserted here.

2 A weak energy certificate with all boundary data retained

Let U be as in (3), and let $\omega_0 \in L^2(U;\mathbb{R}^2)$ represent the prescribed affine class. Set $V = H^1(U)/\mathbb{R}$ with $\|h\|_V = \|\nabla h\|_2$ and $\mathcal{G} = \nabla H^1(U) \subset L^2(U;\mathbb{R}^2)$. Poincare's inequality makes V a Hilbert space and $\mathcal{G}$ closed. Let P be the orthogonal projection onto $\mathcal{G}$.

Proposition 2.1 (Residual and flux enclosures)**.** *There is a unique minimizing gradient, with* $\omega_* = (I-P)\omega_0$ *and* $\mathcal{M}_U = \|\omega_*\|_2^2$*. For every trial field* $v = \omega_0 + \nabla\widetilde{g}$*, define* $\ell_v(h) = \int_U v\cdot\nabla h$ *and* $\eta = \|\ell_v\|_{V'}$*. Then*

$$\mathcal{M}_U = \|v\|_2^2 - \eta^2, \qquad \eta = \|Pv\|_2 . \tag{4}$$

If $\eta \le e$ *is certified, then* $\max(0, \|v\|_2^2 - e^2) \le \mathcal{M}_U \le \|v\|_2^2$*. For any* $q \in \mathcal{G}^\perp$*,*

$$\|v-q\|_2^2 = \|v-\omega_*\|_2^2 + \|\omega_* - q\|_2^2 , \tag{5}$$

$$\mathcal{M}_U \ge 2\langle v, q\rangle - \|q\|_2^2 = \|v\|_2^2 - \|v-q\|_2^2 . \tag{6}$$

Proof. The variational equation is $\int \nabla g_*\cdot\nabla h = -\int \omega_0\cdot\nabla h$ for all $h \in V$. Riesz representation gives $\nabla g_* = -P\omega_0$. For any admissible v, $(I-P)v = \omega_*$ and Pv is the Riesz representative of ℓ_v. Orthogonal decomposition proves (4). In (5) the first difference lies in $\mathcal{G}$ and the second in $\mathcal{G}^\perp$. Finally $\langle v, q\rangle = \langle \omega_*, q\rangle$, and completing the square proves (6). □

This is the classical projection/hypercircle mechanism, not a new PDE existence theorem. Its source-specific role is to certify exactly the energy appearing in (3). A verified interval $M_- \le \mathcal{M}_U \le M_+$ gives

$$e^{-M_+/(2a)} \le e^{-\mathcal{M}_U/(2a)} \le e^{-M_-/(2a)}. \tag{7}$$

The weak inverse used here is the Neumann inverse on V; the Dirichlet Green function in Section 3 enters through a rotated stream function, not by changing the boundary condition of g_*.

2.1 What a computable residual must include

For sufficiently regular q, membership in $\mathcal{G}^\perp$ means $\operatorname{div} q = 0$ and zero normal trace on *every* boundary component. Divergence alone does not suffice. More generally, if

$$\ell_v(h) = \int_U j\cdot\nabla h + r(h), \qquad \|r\|_{V'} \le e_0, \tag{8}$$

then $\eta \le \|j\|_2 + e_0$. Piecewise fluxes must account for internal normal jumps, boundary traces, and source/quadrature defects in r. Any numerical integration used in an upper bound must itself

be enclosed. A finite set of tested residual values gives a *lower*, not upper, bound on η unless an independent complement estimate is provided.

Period verification is logically prior to residual verification. On a punctured disk the zero field has zero weak residual and zero energy, but is not admissible for nonzero circulation. A finite coordinate Gram matrix must represent the physical L^2 metric; arbitrary coefficient Euclidean norms need not bound the displayed energies.

Fresh original private comparisons were made with [2], lines 53–190, 492–558, 1008–1037, and [3], lines 478–599, 842–932. These passages supply boundary compatibility, physical-metric pullback, flux equilibration, and the distinction between closed and exact fields. They were read as local proof material, not as authority for the XY critical input. Their full unread complements are recorded separately.

3 An exact circular-hole defect and a quadratic enclosure

Fix distinct $z_1, \dots, z_k \in D$, real charges l_i, and radii $\rho_i > 0$ such that the closed disks $\overline{B}(z_i, 2\rho_i)$ are pairwise disjoint and contained in D. Put

$$\Omega = D \setminus \bigcup_i \overline{B}(z_i, \rho_i), \qquad U(x) = \sum_i l_i G_D(x, z_i), \tag{9}$$

$$u_i(x) = U(x) - \frac{l_i}{2\pi} \log \frac{1}{|x - z_i|}, \qquad d_i = \int_{B(z_i,\rho_i)} |\nabla u_i|^2, \quad d = \sum_i d_i, \tag{10}$$

$$E_0 = 2\pi \sum_i l_i^2 \log(1/\rho_i) + 4\pi^2 \left[\sum_i l_i^2 R_D(z_i, z_i) + 2 \sum_{i<j} l_i l_j G_D(z_i, z_j) \right]. \tag{11}$$

Each u_i is harmonic near the larger disk. Let $J(x, y) = (-y, x)$ and $v = -2\pi J \nabla U$. Its counterclockwise periods around the holes are $2\pi l_i$. It is divergence free and tangent to the outer boundary. Let $\mathcal{M}$ be the minimum in its affine class on Ω. The analytic result allows real charges; the source's integer-height sectors still require integer charges, and its annular gluing uses unit charges. Those discrete restrictions are not removed.

Theorem 3.1 (Exact defect and explicit collar certificate)**.** *With P the gradient projection on Ω,*

$$\|v\|_2^2 = E_0 - 4\pi^2 d, \qquad E_0 - \mathcal{M} = 4\pi^2 d + \|Pv\|_2^2 . \tag{12}$$

Moreover,

$$4\pi^2 d \;\le\; E_0 - \mathcal{M} \;\le\; \frac{32\pi^2}{3} d, \qquad d \le \pi \sum_i \rho_i^2 \sup_{B(z_i,\rho_i)} |\nabla u_i|^2. \tag{13}$$

Thus, for a fixed separated configuration and fixed charges, the remainder is one-sided and $O(\sum_i \rho_i^2)$. The source's exterior-energy lemma states $O(\rho)$ for equal radii and unit charges.

Proof of the exact identity. Green's identity gives $\int_\Omega |\nabla U|^2 = \int_{\partial\Omega} U \partial_n U$. The outer contribution vanishes. On the ith inner circle, $n = -e_r$ and

$$U = \frac{l_i}{2\pi} \log(1/\rho_i) + u_i, \qquad \partial_n U = \frac{l_i}{2\pi\rho_i} - \partial_r u_i.$$

The mean-value property and harmonicity give $\int_0^{2\pi} u_i \, \mathrm{d}\theta = 2\pi u_i(z_i)$ and $\int_{\partial B_i} \partial_r u_i = 0$. The remaining term is exactly $\int_{\partial B_i} u_i \partial_r u_i = d_i$. Consequently

$$\int_{\partial B_i} U \partial_n U = \frac{l_i^2}{2\pi} \log(1/\rho_i) + l_i u_i(z_i) - d_i.$$

Summing and using symmetry of G_D proves the first equality of (12). The second is Proposition 2.1. This calculation is valid weakly on the outer square; no smooth outer corners are required. □

Proof of the collar bound. In polar coordinates at z_i write

$$u_i = A_0 + \sum_{m\geq 1} r^m(A_m \cos m\theta + B_m \sin m\theta), \quad d_i = \pi \sum_{m\geq 1} m\rho_i^{2m}(A_m^2 + B_m^2). \tag{14}$$

The normal flux on the inner boundary of Ω is $q_i = -(2\pi/\rho_i)\partial_\theta u_i$. It has mean zero. On $\rho_i < r < 2\rho_i$ solve the harmonic Neumann problem with inner normal flux q_i and outer normal flux zero. For a single inner flux $b_m \cos m\theta$, the solution modulo a constant is

$$\chi_m = C_m(r^m + (2\rho_i)^{2m} r^{-m}) \cos m\theta, \qquad C_m = \frac{b_m \rho_i^{m+1}}{m((2\rho_i)^{2m} - \rho_i^{2m})}. \tag{15}$$

Indeed, $-\partial_r \chi_m(\rho_i) = b_m \cos m\theta$ and $\partial_r \chi_m(2\rho_i) = 0$. Its energy, computed by the inner boundary pairing, is

$$\int |\nabla \chi_m|^2 = \frac{\pi b_m^2 \rho_i^2}{m} \frac{4^m + 1}{4^m - 1}. \tag{16}$$

The sine mode has the same formula and is orthogonal to the cosine mode. For q_i, the sum of squared flux amplitudes in degree m is $(2\pi m \rho_i^{m-1})^2(A_m^2 + B_m^2)$. Hence

$$\int |\nabla \chi_i|^2 = 4\pi^2 \left[\pi \sum_{m\geq 1} m\rho_i^{2m}(A_m^2 + B_m^2) \frac{4^m + 1}{4^m - 1}\right] \leq \frac{20\pi^2}{3} d_i. \tag{17}$$

The series statements follow first for finite sums and then by energy convergence; u_i is harmonic on a neighborhood of the closed disk. Extend $j_i = \nabla \chi_i$ by zero outside its collar and put $j = \sum_i j_i$. Zero outer flux prevents a distributional divergence at the collar seam. Thus $v - j$ has divergence zero and zero normal trace on every wall: $\ell_v(h) = \int j \cdot \nabla h$ for every $h \in H^1(\Omega)$. Disjoint collars yield $\|Pv\|_2^2 \leq \|j\|_2^2 \leq (20\pi^2/3)d$. Combining this with (12) proves the theorem. □

3.1 Sharp information retained, and limits of the improvement

The coefficient $5/3$ in (17) is attained by the first harmonic mode of this particular radius-two collar construction. It is not asserted to be the best global projection constant. A collar with outer radius $\lambda\rho_i$ instead gives the multiplier $\coth(m \log \lambda)$ and the bound $4\pi^2 \coth(\log \lambda) d_i$. Larger collars improve this coefficient only when the geometry permits disjoint collars; they are not free near collisions. The full mode sum in (17) is a more precise certificate than its worst-mode bound.

For a disk of radius R, a single centered charge l has constant regular part u, hence $d = 0$. In this case

$$\mathcal{M} = 2\pi l^2 \log(R/\rho), \qquad e^{-\mathcal{M}/(2a)} = (\rho/R)^{l^2/8} \quad (a = 8\pi).$$

This verifies both the circulation and energy normalizations. For a local regular mode $u = Ar^m \cos m\theta$, the defect is $d = \pi m A^2 \rho^{2m}$, and the collar correction cost is $4\pi^2 d \coth(m \log 2)$. This is an exact local mode calculation, not a claim that every such mode is separately a global square-domain Green function. In particular the first mode displays the quadratic radius scale without an uncontrolled trace constant.

Corollary 3.2 (Sector and polygon consequences)**.** *Under the source's fixed-hole sector-limit hypothesis, the limiting sector ratio F for the circular configuration satisfies*

$$\exp(\pi d/4) \le \frac{F}{\exp[-E_0/(16\pi)]} \le \exp(2\pi d/3). \tag{18}$$

For equal radii, replacing circles by the source's fixed rectilinear shape ρP, where $B(0,1-\varepsilon) \subset P \subset B(0,1+\varepsilon)$, changes its continuum leading-energy approximation by $O(\varepsilon+\rho^2)$, for $0<\varepsilon<1/2$ and fixed separated centers and charges. This improves the corresponding continuum $O(\varepsilon+\rho)$ bookkeeping, not a mesh convergence rate.

Proof. Divide (13) by $2a = 16\pi$ and exponentiate. The ratio itself is also at most one before division by the leading factor, since $\mathcal{M} \ge 0$. For the polygon statement, restriction of an admissible field to the domain with larger holes preserves periods and decreases its integrated energy. Thus the polygon minimum is between the circular minima at radii $\rho(1-\varepsilon)$ and $\rho(1+\varepsilon)$. Their leading logarithms differ from the radius-ρ value by $O(\varepsilon)$; their regular-energy defects are uniformly $O(\rho^2)$ in this fixed configuration. Apply Theorem 3.1 on both sides. □

The order of limits remains essential: geometry and annular parameters are fixed before the mesh limit. Bounds for ∇u_i generally deteriorate when points collide or approach the boundary, or charges grow. The energy certificate alone gives neither an all-order collision majorant nor a uniform lattice error. Center normalization cancels the source's identical microscopic core ratios, not arbitrary continuum residual errors; the latter must still be added with their signs or absolute bounds.

4 A larger stability domain for positive annular kernels

We now isolate exactly the operator lemma used in source Section 5. Let (X_i, m_i) be probability spaces, and let measurable kernels $t_i : X_i \times X_{i+1} \to (0,1]$ satisfy

$$\int_{X_i \times X_{i+1}} (1-t_i)\,\mathrm{d}(m_i \otimes m_{i+1}) \le \xi. \tag{19}$$

The spaces, measures, and kernels may vary with i. Neither symmetry nor row-wise normalization is assumed. Define $L_i h(y) = \int t_i(x,y)h(x)\,\mathrm{d}m_i(x)$. The terminal density from a core $g : X_1 \to (0,1]$ is the normalization of $L_{N-1}\cdots L_1 g$ by its integral.

Lemma 4.1 (Normalization projection)**.** *If $q \in L^2(m)$ on a probability space and $\int q = 1$, then $P_q h = h - q\int h$ satisfies*

$$\|P_q\|_{2\to 2} \le \|q\|_2 . \tag{20}$$

This is equality whenever the mean-zero subspace is nontrivial.

Proof. Write $q = 1 + r$ and $h = b1 + z$, with $\int r = \int z = 0$. Then $P_q h = z - br$ and $\|z-br\|_2^2 \le (1+\|r\|_2^2)(|b|^2 + \|z\|_2^2)$. This proves the bound because $\|q\|_2^2 = 1 + \|r\|_2^2$. When $r \ne 0$, the two-dimensional span of $1, r$ gives equality by the norm of the row operator $(b,z) \mapsto z - br$. When $r = 0$, any nonzero mean-zero z gives equality. The one-dimensional probability space is the harmless exception, where $P_q = 0$. □

Theorem 4.2 (Improved average-loss threshold)**.** *Assume (19) with*

$$0 < \xi < \sqrt{5} - 2, \qquad B = \frac{2}{1+\sqrt{1-4\xi}}, \qquad \rho = \sqrt{\xi}\,B^{3/2} < 1. \tag{21}$$

Choose any $\delta > 0$ such that $m_1\{g < \delta\} < 1/(2B)$. For $N \geq 2$ the terminal normalized density p_N obeys

$$p_N \leq B + \frac{2}{\delta} B(1 + B\sqrt{\xi})\rho^{N-2} \left\| g - \int g \right\|_{L^2(m_1)} \quad a.e. \tag{22}$$

In particular $p_N \leq 2$ for all sufficiently large N. The bound is uniform over kernels satisfying the assumptions for this ξ and g. No uniform depth over all positive g is asserted.

Proof. The smaller quadratic root B satisfies

$$\xi B^2 = B - 1, \qquad (1 - B\xi)^{-1} = B. \tag{23}$$

Start $q_1 = 1$, and recursively put

$$\beta_i = \int L_i q_i \, \mathrm{d}m_{i+1}, \qquad q_{i+1} = \beta_i^{-1} L_i q_i.$$

If $0 \leq q_i \leq B$ and $\int q_i = 1$, then $L_i q_i \leq 1$ and $\beta_i \geq 1 - B\xi = 1/B$. Induction therefore gives

$$\int q_i = 1, \quad 0 \leq q_i \leq B, \quad \|q_i\|_2 \leq \sqrt{B}, \quad \beta_i^{-1} \leq B. \tag{24}$$

Using these same β_i, define $f_1 = g$, $f_{i+1} = L_i f_i/\beta_i$, $c_i = \int f_i$, and $E_i = f_i - c_i q_i$. Thus $\int E_i = 0$ and

$$E_{i+1} = \beta_i^{-1} P_{q_{i+1}} L_i E_i. \tag{25}$$

For mean-zero E, $L_i E = (t_i - 1)E$. Since $0 \leq 1 - t_i \leq 1$, the Hilbert–Schmidt estimate gives

$$\|L_i E\|_2 \leq \sqrt{\xi} \, \|E\|_2 . \tag{26}$$

Together with Lemma 4.1 and (24), this yields $\|E_{i+1}\|_2 \leq \rho \, \|E_i\|_2$. Moreover $\rho^2 = \xi B^3 = B(B-1) < 1$ precisely when $B < (1 + \sqrt{5})/2$, which is equivalent on the smaller-root branch to $\xi < \sqrt{5} - 2$. This establishes contraction without the source's cruder factor from normalization.

We must also keep c_i away from zero. Express it by reversing the kernel string:

$$c_i = \frac{\int g(x_1) \prod_{j<i} t_j(x_j, x_{j+1}) \prod_{j \leq i} \mathrm{d}m_j(x_j)}{\int \prod_{j<i} t_j(x_j, x_{j+1}) \prod_{j \leq i} \mathrm{d}m_j(x_j)}.$$

Backward normalization starting from the terminal function one satisfies exactly the same induction (24), since the loss assumption is with respect to the product measures. The resulting density w_i on X_1 has integral one and is at most B. Consequently

$$c_i = \int g w_i \, \mathrm{d}m_1 \geq \delta(1 - B m_1\{g < \delta\}) > \delta/2. \tag{27}$$

For $i = 1$ use $w_1 = 1$. Existence of such a δ follows from $g > 0$ almost everywhere; no quantitative lower bound on g is assumed.

Finally, $\|L_i E\|_\infty \leq \|E\|_2$ by $t_i \leq 1$ and Cauchy–Schwarz, while $|\int L_i E| \leq \sqrt{\xi} \, \|E\|_2$ by (26). Therefore

$$\|E_{i+1}\|_\infty \leq B(1 + B\sqrt{\xi}) \, \|E_i\|_2 .$$

Since $E_1 = g - \int g$ and $p_N = f_N/c_N = q_N + E_N/c_N$, (27) and the L^2 contraction imply (22). As $B < (1 + \sqrt{5})/2 < 2$, its remainder eventually fits below $2 - B$. □

4.1 Depth, an explicit example, and scope

Let $A = (2/\delta)B(1 + B\sqrt{\xi})\,\|g - \int g\|_2$. If $A > 0$, a sufficient integer depth is

$$N \geq 2 + \max\left\{0, \left\lceil \frac{\log(A/(2-B))}{\log(1/\rho)} \right\rceil \right\}. \tag{28}$$

If g is constant, $A = 0$ and every $N \geq 2$ works. For a concrete nontrivial parameter choice, $B = 3/2$ gives $\xi = 2/9$ and $\rho^2 = 3/4$, already well beyond $1/64$. If $g \geq 1/4$ almost everywhere, take $\delta = 1/4$. Since $0 < g \leq 1$, its variance is at most $1/4$; thus $A \leq 6(1 + 1/\sqrt{2}) < 12$. Taking $N = 26$ suffices, because $(\sqrt{3}/2)^{24} = (3/4)^{12} < 1/24$, so $A\rho^{24} < 1/2 = 2 - B$. This example does not replace the source's potentially very small core-dependent δ by a universal constant.

The case $\xi = 0$ is simpler: $t_i = 1$ almost everywhere and the terminal density is one after a single step. At $\xi = \sqrt{5} - 2$ our contraction factor is one. Failure of this proof at that endpoint is not an impossibility theorem or a claim that the threshold is optimal among all methods. The improvement is relative to the source's stated stability lemma, not a historical priority claim for positive kernels.

The joint exterior attachment in the source can be tested against the product of terminal nest densities. If each is at most $B + e_N$, a nonnegative exterior loss with integral at most ξ_{ext} under the cut product measure has expectation at most $(B + e_N)^k \xi_{\text{ext}}$. This is only a change-of-density estimate; it does not assert independence inside the exterior. Here k is fixed before choosing losses and depths. Verification that the actual XY kernels have (19), and that the core is positive, remains part of the source's sector, annular, and pinning arguments.

5 Comparison, advantages, and remaining obligations

Interface	Source statement	Local result and restriction
Circular energy	Equal radii, unit charges, $O(\rho)$ remainder	Exact negative regular-energy defect; explicit $O(\sum \rho_i^2)$ enclosure for fixed real charges and separated disks.
Kernel propagation	Average loss $\xi < 1/64$, eventual density at most two	$\xi < \sqrt{5} - 2$, finite-depth bound on varying spaces; core-dependent δ remains.
Approximate PDE energy	Exact variational minimum used in sector limit	Primal/flux enclosure with all normal traces and periods; requires a verified full dual residual, not sampled residuals alone.
Critical spin field	Full-sequence normalized imaginary-chaos limit	Attributed to source; no independent certification of the complete critical-height, collision, and tightness chain.

The circular improvement removes an avoidable first-order remainder without requiring additional lattice information. Its cost is that the computable d_i involve the regular Green field, or an independently certified upper bound for it. The collar certificate is explicit but may be conservative because it does not solve the global projection problem. It is especially useful for unequal hole radii and certified continuum experiments at a fixed configuration. It does not control collisions.

The kernel argument tolerates a substantially larger average loss without requiring a pointwise lower bound on t_i. Its cost is that the depth depends on the lower tail of g and deteriorates near the displayed threshold. Neither result turns an abstract transform or finite Gram calculation into a proof of critical stiffness. The private residual and metric machinery makes these distinctions explicit; its classical components are not counted as the principal contribution.

Useful next steps are certified regular-Green derivative bounds on specified separated configurations, comparison of the explicit collar flux with a global equilibrated flux, and sharper kernel propagation using additional information actually available from the annuli. A collision-uniform energy theorem, an effective lattice-to-sector rate, or a stronger topology for a mollified lattice field would each require new estimates and is not established here. The optional iteration of localization buffers is not needed or asserted in this report.

6 Current public literature and evidence boundaries

Public primary abstract and metadata checks were refreshed in this authoring stage. Lammers' *Bijecting the BKT transition* [5] relates spin and height correlation lengths. The 2026 preprint of Durand and Lammers [4] relates XY mass to a one-sided surface-tension derivative, with a quadratic error near the origin. These abstracts do not assert the critical spin-field limit being compared here, nor do they independently supply its exact critical curvature. Van Engelenburg and Lis [6] provide a phase-transition result; this is not the same as identifying the critical field.

Junnila, Saksman, and Webb [7] study imaginary multiplicative chaos, its moments and regularity. Their construction is relevant continuum context, not identification of this particular lattice XY limit. Only these public records were sent to public search. No private report text or private title was uploaded. The present literature check covers the specified abstracts and metadata, not those papers' complete proofs, and does not establish historical novelty of our local inequalities.

6.1 Literal scope and independent checks

The owned read ledger records all 3,366 lines of the twelve cached public texts, including 3,054 scientific section lines, plus 448 lines of the two original private reports specified above. Public reference files were read, but reading their entries does not mean reading the cited works. Pre-read hashes were persisted before these literal displays; post-read hashes and unread complements are recorded in the final evidence. The parent's captured-tree identity is provenance, not live-branch status or proof certification. No parent test count is included in our counts.

The accompanying executable uses exact rational arithmetic for harmonic mode fluxes and collar energies, physical-metric projection identities, normalization projections, and finite nonstationary positive-kernel steps. Its case and assertion totals are recorded in the owned JSON. The executed suite passed 3,285 parameter cases and 22,135 assertions, including 960 kernel steps in 120 chains. All these steps have a pointwise loss somewhere exceeding ξ, while their product-measure average loss is at most ξ. Thus these are not merely fixtures for a stronger pointwise-loss hypothesis. The mode, unequal-hole, weighted hypercircle, oblique-projection, threshold, and depth groups contain 648, 100, 735, 1,620, 61, and 1 cases, respectively. These bounded fixtures detect normalization and sign mistakes; they do not substitute for the infinite-dimensional proofs above or validate the source's complete scaling limit. No foreign checker was executed.

7 Conclusion

The source's circular Green calculation contains more information than its stated $O(\rho)$ remainder: exact harmonic averaging, followed by an explicit collar flux, gives a signed quadratic enclosure. Its kernel stability argument likewise admits a larger quantitative domain when normalization is treated as an oblique projection in the actual probability-space metric. These are the two demonstrated C3-level local increments. Applicability L3 is unchanged and does not determine that

score. All source-wide critical, discrete, and probabilistic dependencies remain separated from these deterministic refinements.

References

[1] OpenAI mathematics manuscript, *The Critical Spin Field of the Planar XY Model*, October 5, 2026. Pinned cached source, especially Sections 3, 5, and 6. Public source snapshot.

[2] M. Subbarao, private original technical report, *TR8-lin-non-lin*, locally supplied source. Fresh proof scope: lines 53–190, 492–558, 1008–1037.

[3] M. Subbarao, private original technical report, *TR-rev-1-old-tr250*, locally supplied source. Fresh proof scope: lines 478–599, 842–932.

[4] T. Durand and P. Lammers, *The BKT transition and surface tension differentiability*, 2026. arXiv:2605.28473.

[5] P. Lammers, *Bijecting the BKT transition*, 2023. arXiv:2301.06905.

[6] D. van Engelenburg and M. Lis, *An elementary proof of phase transition in the planar XY model*, 2021 preprint, published 2023. arXiv:2110.09465.

[7] J. Junnila, E. Saksman, and C. Webb, *Imaginary multiplicative chaos: Moments, regularity and connections to the Ising model*, Ann. Appl. Probab. 30 (2020), 2099–2164. arXiv:1806.02118.